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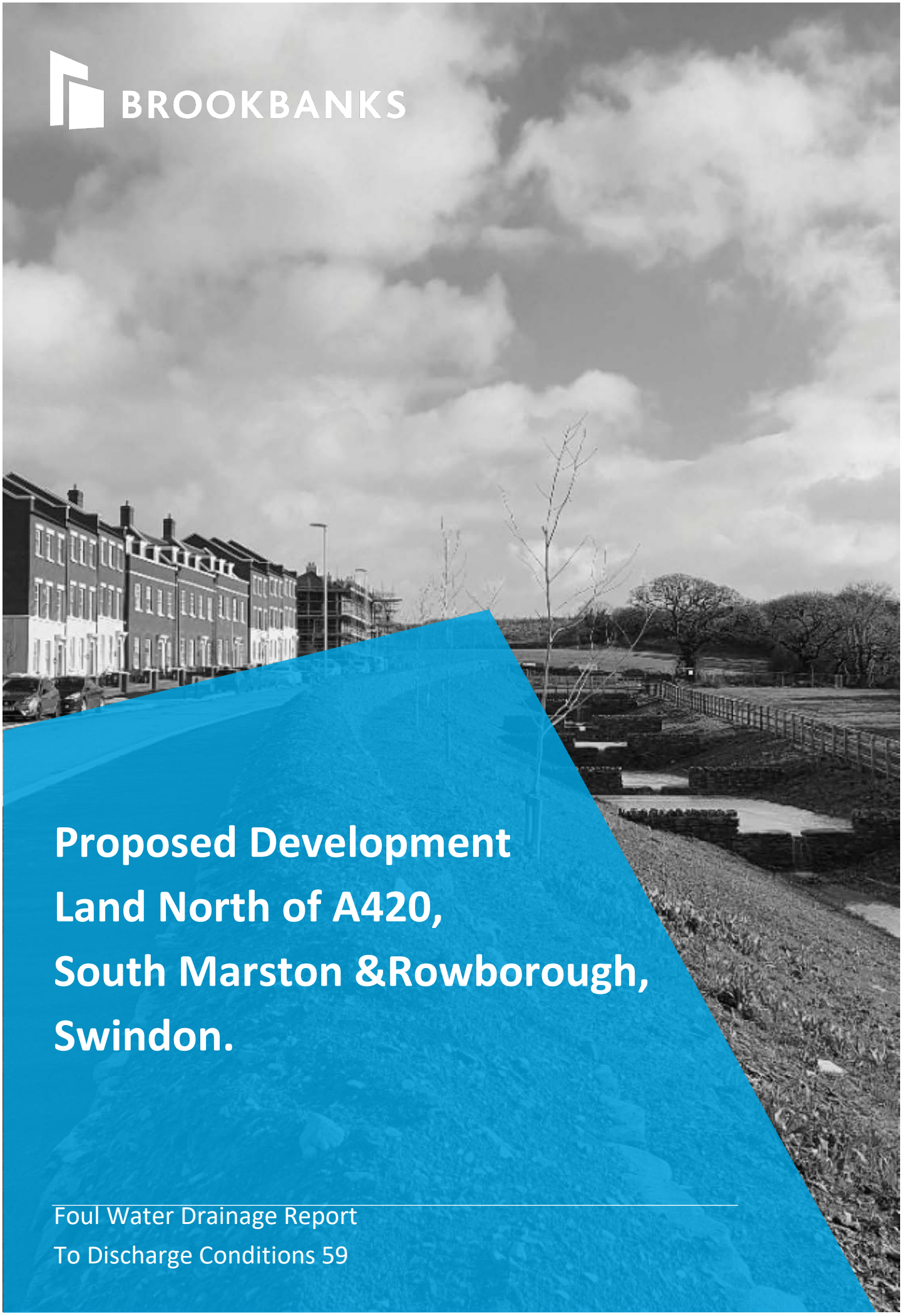

Proposed Development Land North of A420, South Marston &Rowborough, Swindon.

Foul Water Drainage Report

To Discharge Conditions 59

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Appendix E – Phasing Plan LHL005/186/Rev D (sourced by DLA)

1 Introduction

1.1 Brookbanks Consulting (BCL) is appointed by the Swindon New Eastern Villages Consortium to produce a foul water drainage report for a residential-led scheme development at South Marston and Rowborough to the east of Swindon, together with areas of mixed use comprising: community; employment; education; local retail uses; and formal and informal open space provision on land east of Swindon and north of A420.

1.2 This Report has been prepared in order to satisfy the following foul water drainage planning condition with respect to the outline planning application (ref. S/OUT/13/1555).

1.3 Condition 59. Thames Water - Waste Water

Prior to the commencement of the dwellings, commercial and community uses hereby approved, a foul drainage strategy detailing any on and/or off site drainage works shall be submitted to and approved in writing by the Local Planning Authority.

No discharge of foul or surface water from the site shall be accepted into the public system until the drainage works referred to in the strategy, including any interim arrangements, have been completed in accordance with the approved drainage strategy and retained in the approved form.

Reason: The development may lead to sewage flooding; to ensure that sufficient capacity is made available to cope with the new development; and in order to avoid adverse environmental impact upon the community.

1.3 This Report should be read in accordance with the engineering drawings and other documents included in the Reserved Matters submission.

2 Foul Water Drainage

Background

2.1 The strategic foul water drainage scheme is based on the 'Foul Water Drainage Strategy' included in the Flood Risk Assessment (FRA) (ref. 1335/FRA/03) submitted and approved as part of the outline planning application

2.2 Development flows will be collected at two proposed on-site pumping stations and delivered to a third proposed off-site Strategic Pump Station (SPS) to be constructed by Thames Water off the A420 to the south west of the site at Symmetry Park.

2.3 Drainage proposals for individual parcel developments will be subject to separate reserved matters applications which will rely on the strategic foul water scheme outlined in this report.

Baseline Conditions

2.4 The proposed development lies to the east of Swindon being part of the New Eastern Villages, a key strategic allocation in the adopted Swindon Local Plan.

2.5 The A420 and the Great Western Rail Line bound the site to the south. To the west of the site, Thornhill Lane forms the site boundary, with the extents of South Marston, Nightingale Lane and the accompanying farmland bounding the site to the north. Agricultural land forms the eastern boundary of the site.

2.6 The majority of the land is thought to be historically undeveloped, however, small developments including a combination of dwellings, hotels and public houses are spread around the site, with link roads included throughout. The site location and boundary is shown indicatively on Figure 2a, below.

2.7 Development Criteria

The 165 ha site has outline planning permission for:

- Up to 2,380 residential units
- Up to 1,500m^2 Retail Space
- Up to 1,000m^2 Employment Space

Sewer Systems

2.8 A Drainage Strategy Plan – Drawing 1335-SK-115, as included in Appendix C, has been developed based on existing site topography and the Regulating Plan. This indicates two foul drainage catchments covering the development

FWPS	**Catchment Area**	Houses	**Peak Flows(l/s)**
1	R1,R2,R3,R4,R5,R6,R7,R8, R9, R10 R11, R12, R13, R14, R15, R16,R17 (part), R18 (part), Primary School, Mixed Use & Retained Development	1500	69.4 + non-residential flows
2	R17 (part), R18 (part), R19, R20, R21 & R22	957	44.3

Table 3a: Foul catchment areas

2.9 Strategic foul water sewer systems and pumping stations will be designed and constructed in accordance with the Sewerage Sector Guidance Appendix C – Design and Construction Guidance and Thames Water's requirements.

2.10 The strategic foul water systems will incorporate lateral connections at suitable locations to serve all individual development parcels. In accordance with the planning condition, foul drainage schemes for each phase/ sub-phase will be delivered based on this overall strategy.

2.11 FWPS1 serves the majority of the site and is located adjacent to Old Vicarage Lane while FWPS2 serves the eastern section of the site. The strategy proposes that FWPS1 will have adequate capacity for the whole site, as flows from FWPS2 will eventually be pumped westwards to discharge to the head of the proposed gravity foul water system draining to FWPS1.

2.12 The foul water drainage system will be offered for adoption under a section 104 agreement with Thames Water (TW) or a New Appointment company (NAV).

Point of Connection

2.13 Foul drainage proposals have been developed in consultation with TW that provide for an off-site connection to the existing public sewer network. The point of connection for flows pumped from FWPS1

within the site will be to a proposed Strategic Pump Station (SPS) off the A420 to the south west of the site at Symmetry Park.

2.14 TW have submitted a planning application for the land on which the SPS will be located and it is anticipated that the SPS will be operational in 2023. If the residential development commences prior to completion of the SPS, TW have agreed to re-imburse the developers' costs for the tankering of sewage off site.

2.15 The route of the rising main from FWPS1 to the SPS follows mainly within public owned land, however there is a requirement to cross a stretch of third party land. TW have agreed to use their statutory powers to construct the the rising main within the area of third party land under Section 98 of the Water Act. Correspondence with TW is included in Appendix D.

2.16 It has been agreed with TW that the on-site pumping stations will provide for an additional 100l/dwelling of emergency storage over and above the DCG requirement of 160l/dwelling. Correspondence with TW is included in Appendix D.

2.17 It is considered therefore that the foul drainage proposals will be compliant with the relevant planning condition and FRA objectives for foul water drainage.

Implementation

2.18 Phasing Plan LHL005/186/Rev D showing the four phases of development is included in Appendix E. Particulars of the main foul water infrastructure delivered within each phase are set out below:

Phase 1 – South Marston East

- Foul water pumping station FWPS1 and rising main to off-site Thames Water SPS. Main foul infrastructure and provision for future connections from phases 2 and 4.
- The drainage infrastructure will be delivered in parallel with the construction of the development parcel that it serves and will be completed prior to occupation within each parcel.

Phase 2 – Rowborough West

- Foul water pumping station FWPS2 and rising main to head of gravity system draining to FWPS1. Main foul infrastructure and provision for future connection from phase 3.
- The drainage infrastructure will be delivered in parallel with the construction of the development parcel that it serves and will be completed prior to occupation within each parcel.

Phase 3 – Rowborough East

- Connections to main foul infrastructure draining to FWPS2.
- The drainage infrastructure will be delivered in parallel with the construction of the development parcel that it serves and will be completed prior to occupation within each parcel.

Phase 4 – South Marston West

- Connections to main foul infrastructure draining to FWPS1 and main surface infrastructure draining to detention basin A.
- The drainage infrastructure will be delivered in parallel with the construction of the development parcel that it serves and will be completed prior to occupation within each parcel.

Summary

2.19 A strategy for foul drainage at the site has been developed to meet both national and local policy. Foul drainage proposals have been developed in consultation with TW that provide for an off-site connection to the existing public sewer network.

2.20 The foul water system has been designed so that it is suitable for adoption by TW for future maintenance.

2.21 Implementation of the foul water management system for each development parcel will be undertaken prior to first occupation of dwellings.

3 Next Steps and Recommendations

3.1 The purpose of this Report is to set out details of the proposed foul and surface water drainage schemes to manage foul flows and surface water run-off from the development, as required by Condition 41 and 59 of the outline planning permission.

3.2 This Report demonstrates that against the requirements of the relevant condition a suitable scheme for the discharge of foul water flows has been designed in accordance with the approved Foul Water Drainage Strategy contained within the Flood Risk Assessment (FRA) (ref. 1335/FRA/03). The drainage scheme, will be designed and constructed in accordance with DCG standards with outfalls at points of discharge agreed with TW. The impacts of development during the construction and operational stages will be mitigated in accordance with agreements with TW.

4 Limitations

4.1 The conclusions and recommendations contained herein are limited to those given the general availability of background information and the planned usage of the site.

4.2 Third party information has been used in the preparation of this report, which Brookbanks Consulting Ltd, by necessity assumes is correct at the time of writing. While all reasonable checks have been made on data sources and the accuracy of data, Brookbanks Consulting Ltd accepts no liability for same.

4.3 The benefits of this report are provided solely to the members of the Swindon Eastern Villages Consortium for the proposed development only.

4.4 Brookbanks Consulting Ltd excludes third party rights for the information contained in the report.

Appendix A

Appendix B

Appendix C

Appendix D

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